KEEPING LLMS FOCUSED: DELAYING ERROR AVALANCHES VIA STRUCTURAL BOUNDARIES

ABSTRACT

Large language models exhibit sharp accuracy declines on multi-step deterministic reasoning tasks even at very short sequence length, often attributed to attention-driven error cascades. Existing mitigations rely on multi-call "divide-and-conquer" strategies. We propose a lightweight alternative: inserting structural boundary in the input. We show that increasing boundary density linearly improves sequence survival rates and delays the accuracy cliff. Analysis of attention patterns reveals that boundary tokens act as computational sinks, absorbing off-diagonal attention at intermediate layers. By periodically interrupting long-range attention dependencies, these structural anchors suppress error propagation, providing a simple structural mechanism for improving reliability in long-context generation.

***K*eywords** Large Language Models · Deterministic Reasoning · Error Cascades

1 Introduction

Evaluating large language models (LLMs) on deterministic, multi-step reasoning tasks reveals critical vulnerabilities. Many domain such as scientific and algebraic domains require exact correctness, where a single sequence-level failure denotes complete task failure. Recent evaluations of deterministic sequence prediction establish that LLMs suffer from an accuracy cliff. Rather than an expected exponential decay in success rates, the Sequence Accuracy Rate (SAR) exhibits a sharp, double-exponential drop beyond a characteristic length scale [1].

This degradation suggests that LLMs fail to execute sequential operations independently. Previous work modeled this phenomenon as a Sherrington-Kirkpatrick (SK) spin-glass, wherein all-to-all attention mechanisms induce destructive interference among generated tokens [1]. To mitigate this correlated error propagation, dividing the problem into sub-tasks via multiple independent API calls has been shown to artificially cut correlation loops. However, this multi-call "divide-and-conquer" strategy introduces computational overhead and formatting bottlenecks.

In this work, we propose a single-pass mitigation strategy: the insertion of Structural Boundaries. By introducing specific boundary tokens (e.g., vertical bars "|") at fixed intervals, we manipulate the LLM's internal error cascade without requiring multiple API calls. We demonstrate that these boundaries act as attention anchors, breaking the attention-induced correlation loops and isolating token generation into conditionally more independent blocks. Through evaluation on deterministic cyclic shift tasks, we show that tuning the frequency of these boundaries significantly delays the accuracy cliff enabling the model to recover from localized errors rather than cascading into full failure.

2 Methods

2.1 Synthetic Task Design

To study boundary sensitivity in large language models, we construct a controlled symbolic transformation task with deterministic rules. This design allows us to systematically manipulate sequence structure while keeping the underlying transformation simple.

2.1.1 Base Task: Cyclic Shift

Let the alphabet be defined as

$$\mathcal{A} = \{A, B, C, \dots\}, \tag{1}$$

with cardinality $|\mathcal{A}| = A$.

An input sequence of length N is sampled independently and uniformly from the alphabet:

$$x = (x_1, x_2, \dots, x_N), \qquad x_i \sim \text{Uniform}(\mathcal{A}). \tag{2}$$

The task is to apply a deterministic cyclic shift to each token:

$$f(x_i) = \text{next}(x_i) \pmod{|\mathcal{A}|}. \tag{3}$$

For example, when the alphabet size is $A = 4$, the transformation is

$$A \to B \tag{4}$$
$$B \to C \tag{5}$$
$$C \to D \tag{6}$$
$$D \to A. \tag{7}$$

Thus, the model must transform each symbol independently according to this rule while preserving the sequence structure.

2.1.2 Cellular Boundary Task

To examine whether structural delimiters induce locality effects, we introduce periodic boundary markers (|) into the sequence. Specifically, a delimiter is inserted after every k tokens:

$$x = x_1 \dots x_k \mid x_{k+1} \dots x_{2k} \mid \dots \tag{8}$$

The delimiter functions as a structural marker: it must be copied exactly and should not undergo the cyclic transformation. All other tokens remain subject to the same shift rule defined above.

This design allows us to test whether explicit boundaries alter the model's ability to perform token-level transformations over long sequences.

2.2 Experimental Grid

We evaluate the model across a full factorial grid of task parameters:

- Alphabet size: $A \in \{2, 4, 8\}$
- Sequence length: $N \in \{32, 64, 128\}$
- Boundary interval: $k \in \{0, 2, 4, 8, 16\}$ (with $k = 0$ indicating no delimiters)
- Trials per configuration: $T = 10$

This experimental design enables us to examine how task complexity, sequence length, and structural segmentation affect model performance. For reproducibility, Appendix A provides the prompt templates used in our experiments.

3 Result

3.1 Scaling Behaviour of Boundary Frequency

To quantify the efficacy of structural interventions, we evaluated the Total Token Accuracy (TAR) as a function of boundary frequency, defined as the inverse of the boundary interval, $1/k$, where $1/k = 0$ corresponds to baseline without boundaries. Results reported in this subseciton use `Gemini Flash 2.5` as the base model. For comparison with an open-source alternative, see Appendix B, which presents results obtained using `Qwen2.5-7B-Instruct`.

As illustrated in Figure 1, TAR exhibits a linear recovery as the density of structural boundaries increases. This positive linear trend persists across different sequence lengths N and alphabet complexities A, indicating the presence of a scaling behaviour governing boundary-induced error mitigation.

Accuracy rises predictably from the random-chance baseline of $1/A$ toward perfect sequence reconstruction as $1/k$ increases. To validate this observation, we performed linear regression analyses for all configurations. The resulting goodness-of-fit (R^2) statistics confirm a significant linear correlation between boundary frequency and token accuracy. This confirms the hypothesis that the probability of sequence survival scales directly and proportionally with the frequency of attention-disrupting structural anchors. The model's deterministic reliability becomes a predictable linear function of boundary density.

We can formally express this relationship as

$$\text{TAR}(k) \approx \frac{1}{|\mathcal{A}|} + \beta \left(\frac{1}{k}\right)$$

where β is a proportionality constant reflecting the model's inherent baseline capability and the complexity of the task, here the alphabet $\mathcal{A}$.

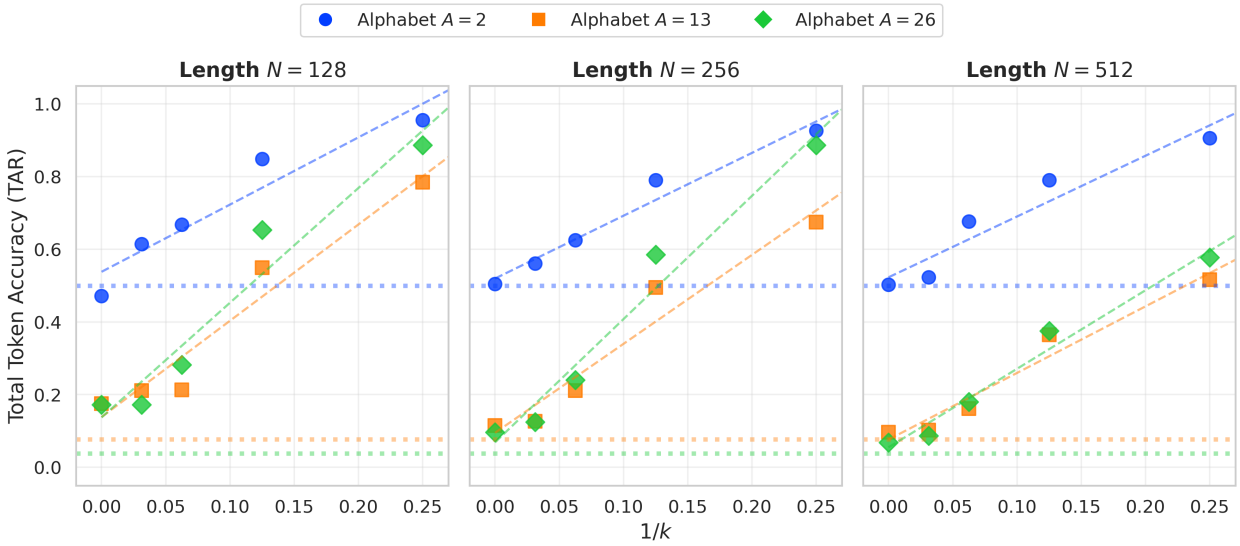


Figure 1: Total Token Accuracy (TAR) as a function of boundary frequency ($1/k$) for various sequence lengths N and alphabet sizes A. Dotted horizontal lines indicate the theoretical random-chance baseline $1/A$.

Table 1: Linear regression goodness-of-fit (R^2) for TAR vs. boundary frequency ($1/k$) across sequence lengths N and alphabet sizes A.

Sequence Length N	$A = 2$	$A = 13$	$A = 26$
128	0.910	0.944	0.941
256	0.966	0.945	0.969
512	0.915	0.956	0.976

3.2 Mechanistic Interpretability

To understand how structural interventions affect the internal representations of large language models, we performed an attention analysis on the open-source model `Qwen2.5-7B-Instruct`. For this analysis, chosen parameters are $A = 2$, $k = 2$,and $N = 16$, respectively. We chose these parameters because the performance of `Qwen2.5-7B-Instruct` was roughly ideal for this setting, achieving an accuracy of approximately 90% with boundaries and 50% without them. Two conditions were compared: a *Base Task* with no structural boundaries and a *Cellular Task* where boundaries were inserted. Figure 2 illustrates an example attention matrix for the Base and Cellular models at a specific layer and head.

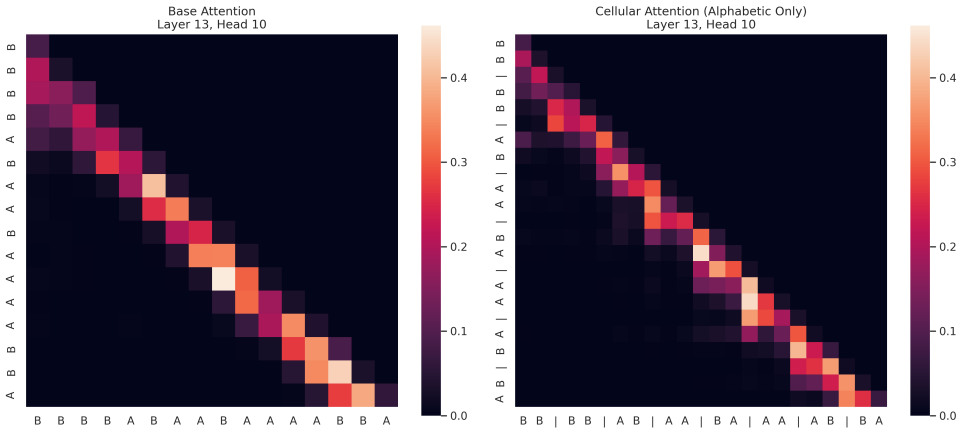


Figure 2: Example attention matrices for the Base (left) and Cellular (right) models at Layer 13, Head 10. Each heatmap shows the attention weights between tokens in the input sequence.

3.2.1 Boundaries as Computational Sinks

We extracted the attention tensors across all 28 layers and 28 heads of the `Qwen2.5-7B-Instruct` model. Specifically, we isolated the attention mass directed explicitly at the inserted boundary tokens. As visualized in Figure 3, the boundary tokens do not act as passive text; rather, they serve as computational sinks for the model's attention mechanism. While early layers distribute attention broadly, the boundary attention mass spikes in the middle layers. Specifically, the boundary absorption peaks at Layer 13, capturing 19.3% of the total attention mass, with secondary spikes at Layers 15, 18, and 20.

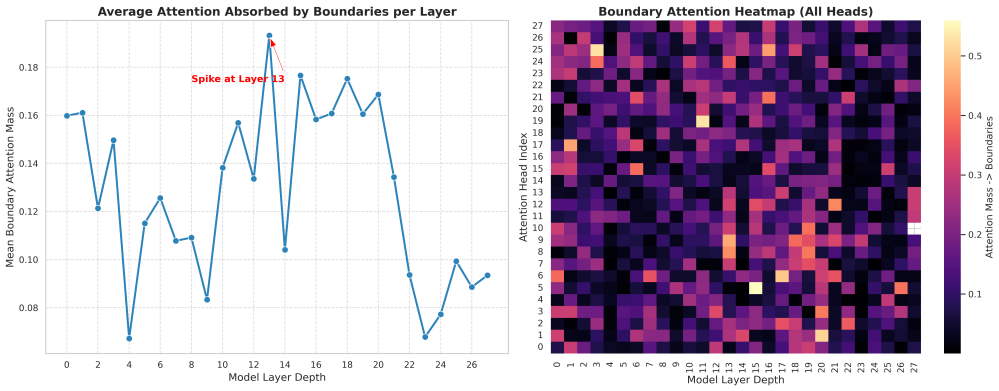


Figure 3:

By absorbing the attention probability mass at these layers, the structural boundaries prevent tokens from attending to distant, potentially corrupted tokens earlier in the sequence context. This mechanistic localization potentially provides quantitative proof for our behavioral observations: structural boundaries artificially isolate sequence generation into conditionally more independent blocks.

3.2.2 Entropy of Attention Heads

We calculated the Shannon entropy of the attention probability distributions ($H = -\sum p \log_2 p$) for every attention head across all 28 layers. We compared the baseline model's entropy against the entropy of the model processing sequences partitioned by structural boundaries, yielding the $\Delta_{Entropy}$ metric. Figure. 4 corroborates this by showing the global distribution of head entropy. As illustrated in the left panel of Figure 4, the layer-wise mean attention entropy reveals that the Cellular model consistently maintains higher entropy across all layers compared to the Base model. Also the distribution of head entropies shows a rightward shift for the Cellular configuration.

It is important to note that the introduction of structural dividers inherently increases the overall sequence length, which raises the mathematical upper bound for maximum possible entropy. Consequently, while this overall increase suggests

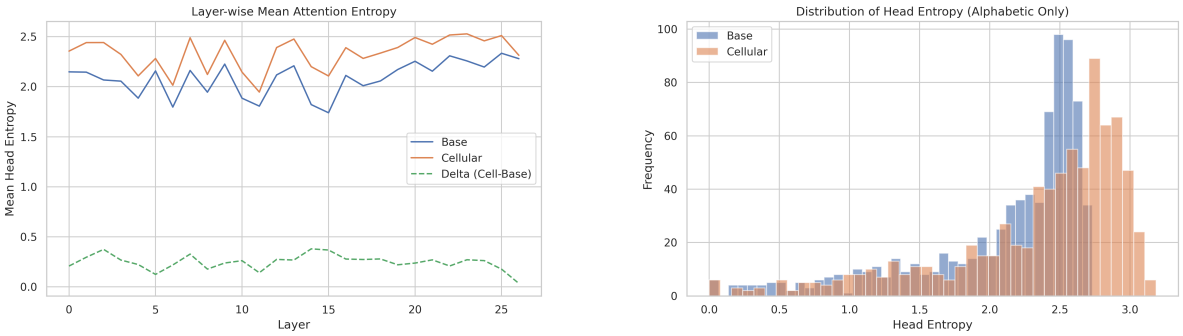


Figure 4: Left: Layer-wise mean attention entropy. Right: Distribution of head entropies.

that the Cellular model's attention heads distribute their probability mass more uniformly across the available context, how this entropy is strategically allocated are investigated further in the subsequent sections.

3.2.3 Analyzing Off-Diagonal Attention Distributions

To quantify the contextual decoupling induced by the structural boundaries, we isolated the off-diagonal elements of the attention tensors. We generated three comparative visualizations to contrast the Base sequences against the boundary-intervened (Cellular) sequences, Figure 5.

The left plot, examines the probability distribution of individual off-diagonal attention weights across all layers and heads, providing a fine-grained view of how attention is allocated between specific token pairs. The center plot shifts the focus to the token level by visualizing the distribution of the total context mass per token, defined as the row-wise sum of all off-diagonal weights; this highlights the aggregate attention each token allocates to its preceding context. Finally, the right plot presents the empirical cumulative distribution function (CDF) of the context mass per token, showing the percentage of tokens in the sequence that allocate at or below a given threshold of total off-diagonal attention.

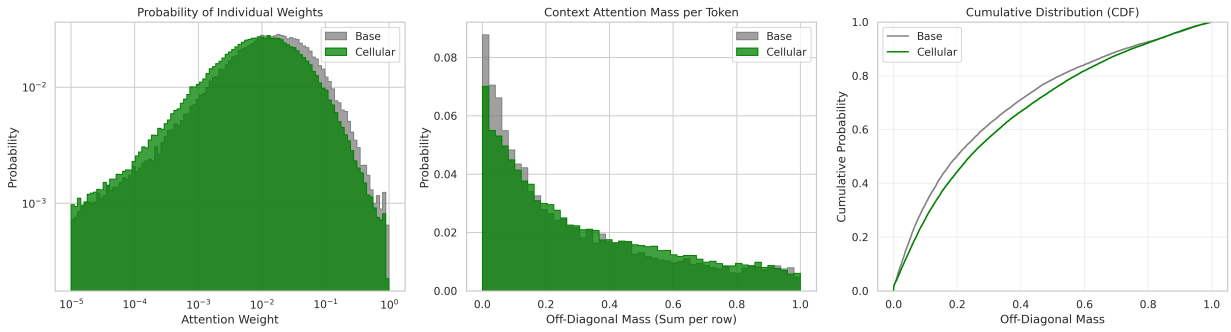


Figure 5: Comparative visualizations of off-diagonal attention distributions. From left to right: Probability of Individual Weights, Probability of Context Mass per Token, and Cumulative Distribution (CDF) of Mass.

While both models exhibit a generally log-normal distribution, the Cellular model demonstrates a distinct leftward shift relative to the Base baseline. The Cellular configuration shows a higher probability of assigning low-magnitude attention weights (the heavier left tail) and a reduced likelihood of assigning high-magnitude weights (the reduced right tail). Our distance-based attention profile, see next section, shows that this shift does not imply uniformly diffused attention, but rather a structured periodic mechanism. The reduction in high-magnitude weights stems from the model offloading direct, long-range token-to-token dependencies onto structural divider tokens.

3.2.4 Distance-Decay of Attention: Periodic Fracturing of Correlation Loops

To observe *where* and *how* the attention-induced correlation are severed, we analyzed the distance-decay profile of the model's attention mechanism. We computed the mean attention weight as a function of the positional distance between the token i and the historical context token j (Distance = $i - j$). These values were averaged across all layers and heads. We decomposed the Cellular model's attention profile into three distinct relational pathways: global attention, attention

directed strictly at the boundary tokens (To Divider), and attention between the core reasoning tokens (Alpha-to-Alpha), see Figure. 6.

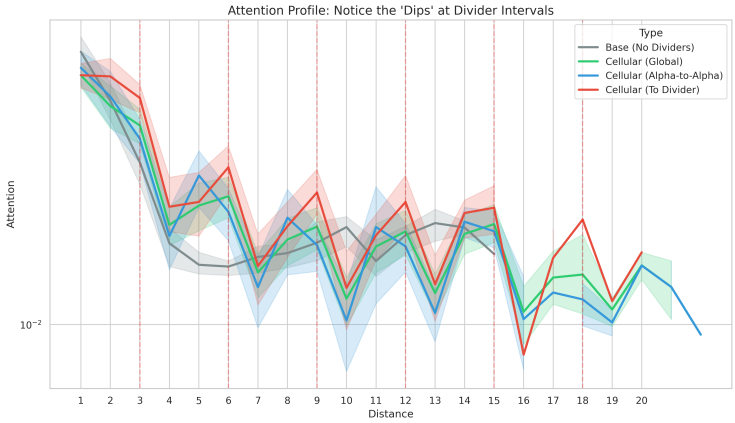


Figure 6: Mean attention weight versus token distance (log scale).

While our initial analysis suggested that the Cellular configuration diffuses attention more broadly, the distance-based attention profile shows a structured, localized periodic mechanism. Unlike the Base model, which exhibits a decay in attention as token distance increases, the Cellular model's attention is modulated by the presence of structural divider tokens. We observe periodic spikes in attention directed toward these dividers at interval boundaries, mirrored by sharp drops in direct letter-to-letter attention at same distances. This indicates that the previously observed leftward shift in overall attention weights is not a flattening of focus; rather, the dividers are acting as attention sinks. The Cellular model chunks the context into distinct, localized cells.

4 Conclusion

In this work, we addressed the accuracy cliff observed in Large Language Models during deterministic, multi-step reasoning tasks. To mitigate this degradation, we introduced a targeted structural intervention: the insertion of boundary tokens. Our empirical evaluations demonstrate that the survival rate of accurate reasoning scales linearly with boundary density, effectively rescuing these models from systemic, length-induced failure.

Through a mechanistic interpretability lens, we revealed how these structural boundaries alter the model's internal routing. Rather than uniformly diffusing attention, these boundary tokens act as localized attention sinks that periodicize the context window into semi-independent cells. This strategy fractures the attention-induced correlation loops that typically drive deterministic error cascades. By offloading long-range dependencies onto these structural hubs, the intervention starves the cascade of the cross-token attention it requires to propagate.

Ultimately, our findings suggest that simple, syntactical structuring can profoundly influence the underlying geometry of attention. By forcing the network to compartmentalize its focus, this cellular approach offers a scalable, compute-efficient pathway for preserving reliability and computational stability in long-context reasoning.

References

[1] Wanda Hou, Leon Zhou, Hong-Ye Hu, Yubei Chen, Yi-Zhuang You, and Xiao-Liang Qi. How focused are llms? a quantitative study via repetitive deterministic prediction tasks. *arXiv preprint arXiv:2511.00763*, 2025.

A Prompt Construction

For a given alphabet size A, the alphabet is defined as the first A uppercase Latin characters:

$$\mathcal{A} = \{A, B, C, \dots\}. \tag{9}$$

For each experiment, a sequence of length N is sampled uniformly from $\mathcal{A}$. From this same underlying sequence we construct two prompts:

- **Base prompt**: the raw sequence is presented without structural boundaries.
- **Cellular prompt**: vertical bars (|) are inserted after every k tokens to create explicit structural boundaries.

Both prompts instruct the model to apply the same cyclic character shift rule, where each symbol is replaced by the next symbol in the alphabet modulo A. The base and cellular prompts are generated from the *same underlying character sequence*. This pairing ensures that performance differences arise solely from the presence of structural boundaries rather than variation in the sampled input.

A simplified prompt template is shown below:

> You are a deterministic sequence transformer.
> Task: Cyclic character shift.
> Alphabet: A, B, C, D.
> Rule: Replace every character with the next one in the alphabet.
> Constraint: Vertical bars (|) are structural boundaries. Do not shift them. Copy them exactly.
> Output only the transformed sequence.
> Input: `A B C A | D A B C`
> Output:

B Scaling Law Analysis for an Open-Source Model

To demonstrate the robustness of the scaling law across different models, we evaluated an open-source 7B model, `Qwen/Qwen2.5-7B-Instruct`, on the same cyclic character shift task. The analysis considered a range of alphabet sizes $A \in \{2, 4, 8\}$ and sequence lengths $N \in \{4, 8, 16, 32\}$.

We computed the Total Token Accuracy (TAR) as a function of boundary frequency $1/k$ and performed linear regression to quantify the relationship. Our primary evaluation metric focuses on the transformed alphabetic sequence rather than delimiter reproduction. Thus, the reported scaling-law results measure whether the model correctly transforms the letter sequence, using boundaries only as structural scaffolds. Table 2 reports the resulting R^2 goodness-of-fit values for sequences with $N \geq 8$, showing a clear linear scaling between TAR and boundary density.

Table 2: R^2 goodness-of-fit for TAR vs. boundary frequency $(1/k)$ for the open-source model `Qwen/Qwen2.5-7B-Instruct`.

Alphabet A	$N = 8$	$N = 16$	$N = 32$
2	0.998	0.979	0.929
4	0.717	0.930	0.799
8	0.855	0.775	0.676

As shown in Table 2, the TAR demonstrates a strong positive correlation with boundary frequency for most sequence lengths and alphabet sizes, confirming that the scaling law generally holds for `Qwen/Qwen2.5-7B-Instruct`. However, for intermediate sequence lengths ($N = 16$) and larger alphabets ($A = 8$), the R^2 values decrease noticeably, indicating that the model exhibits increased sensitivity to task complexity in these configurations.

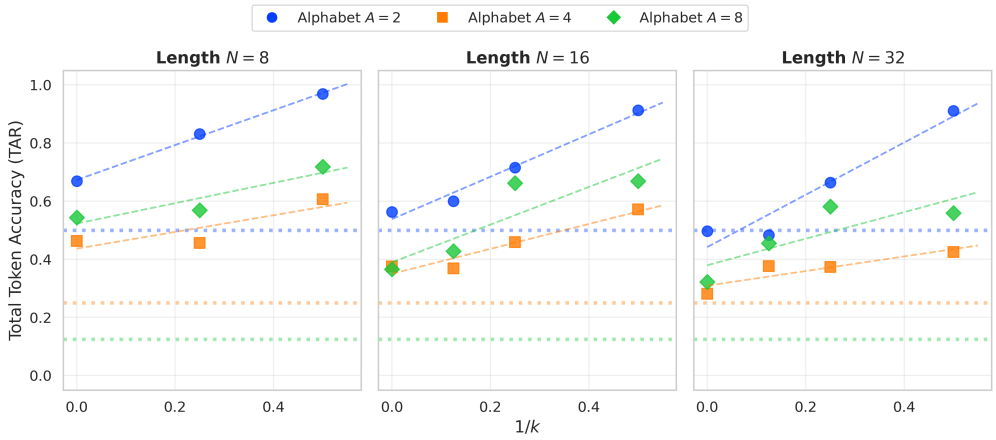


Figure 7: Total Token Accuracy (TAR) as a function of boundary frequency $1/k$ for different alphabet sizes A and sequence lengths N, evaluated on the `Qwen/Qwen2.5-7B-Instruct` model. The dotted horizontal lines indicate the theoretical random-chance baseline $1/A$.